



Clinical implications of the 2021 edition of the WHO classification of central nervous system tumours

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Abstract | A new edition of the WHO classification of tumours of the CNS was published in 2021. Although the previous edition of this classification was published just 5 years earlier, in 2016, rapid advances in our understanding of the molecular underpinnings of CNS tumours, including the diversity of clinically relevant molecular types and subtypes, necessitated a new classification system. Compared with the 2016 scheme, the new classification incorporates even more molecular alterations into the diagnosis of many tumours and reorganizes gliomas into adult-type diffuse gliomas, paediatric-type diffuse low-grade and high-grade gliomas, circumscribed astrocytic gliomas, and ependymal tumours. A number of new entities are incorporated into the 2021 classification, especially tumours that preferentially or exclusively arise in the paediatric population. Such a substantial revision of the WHO scheme will have major implications for the diagnosis and treatment of patients with CNS tumours. In this Perspective, we summarize the main changes in the classification of diffuse and circumscribed gliomas, ependymomas, embryonal tumours and meningiomas, and discuss how each change will influence post-surgical treatment, clinical trial enrolment and cooperative studies. Although the 2021 WHO classification of CNS tumours is a major conceptual advance, its implementation on a routine clinical basis presents some challenges that will require innovative solutions.

Accurate classification of CNS tumours is essential for reliable prognoses and optimal treatment. In 2021 the WHO published the fifth edition of the *WHO Classification of Tumors of the Central Nervous System*^{1,2}. Previous WHO classification schemes relied heavily on histopathological features identified via light microscopy; however, this approach is inherently limited by inter-observer heterogeneity, resulting in relatively high variability in diagnosis and outcomes. The new scheme incorporates advances in the understanding of the molecular pathogenesis of brain tumours, allowing tumours to be grouped into more uniform entities with more predictable natural histories. Thus, the 2021 WHO classification of CNS tumours improves the ability of clinicians to select the most appropriate therapies and enables more homogeneous patient populations to be enrolled into clinical trials, thereby improving the scientific rigour of those trials.

The first internationally accepted classification of CNS tumours was issued in 1979. After the second scheme³ was published 14 years later in 1993, new versions were issued every 7–9 years. Each new version incorporated advances in the understanding of these tumours, including new entities, prognostic subtypes

and terminology. In 2016, a revision of the 2007 fourth edition incorporated molecular criteria into the classification of specific tumour types for the very first time⁴. The newest fifth edition^{1,2} was published in late 2021, representing the shortest interval between editions so far. This shortened time period between editions was the result of a rapid expansion in our knowledge of the molecular underpinnings of CNS tumours, and the realization that molecular alterations are at least as important as histopathological features for accurate diagnosis, prognostic stratification and selection of therapy. In fact, the Consortium to Inform Molecular and Practical Approaches to CNS Tumour Taxonomy–Not Official WHO (cIMPACT-NOW) was created in 2017 as a way to more rapidly disseminate new recommended practices in the pathological work-up of CNS tumours^{5–12} between editions of the WHO classification scheme. A comprehensive discussion of the changes between the 2016 and 2021 editions has already been published¹, so we also encourage review of that article. Here, we focus on how the revisions to the WHO classification scheme will impact the clinical care of patients with CNS tumours, specifically adult-type and paediatric-type diffuse gliomas, circumscribed gliomas, ependymomas, embryonal

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Key points

- The new 2021 WHO classification of CNS tumours has further integrated molecular data into the typing, subtyping and grading of major tumour groups.
- Such integration especially affects the classification of adult-type and paediatric-type diffuse gliomas, circumscribed astrocytic gliomas, ependymomas, embryonal tumours and (to a lesser extent) meningiomas.
- The strengths of this revised scheme include more accurate conceptualization of CNS tumour types, improved diagnostic accuracy and more reliable prognostic subgroups.
- Challenges include greater need for faster, more widespread molecular testing, more issues with third party payor reimbursement, and greater difficulty in finding and enrolling patients who are eligible for specific clinical trials.

WHO grade

World Health Organization tumour grading system; the higher the grade, the greater the tumour malignancy.

Anaplastic

Old term referring to CNS WHO grade 3 tumours.

Chromosome 1p/19q codeletion

Unbalanced translocation leading to a hybrid 1p/19q chromosome that is subsequently lost; one of the hallmark molecular alterations in oligodendrogliomas.

tumours and meningiomas. The updated WHO classification scheme has far-reaching implications for diagnosis, management, the development of national and international therapeutic studies, the rational incorporation of molecular targeted therapies, and, in the near future, the use of combined immunological and molecular approaches to treatment.

General nomenclature

The 2021 WHO CNS tumour classification has expanded on the layered reporting structure that was introduced in the 2016 edition, recommending an integrated diagnosis based on histological features, CNS WHO grade, and molecular information (for example, astrocytoma, WHO grade 4, isocitrate dehydrogenase (IDH)-mutant). Previously, CNS tumour grades were written using Roman numerals. However, in the 2021 WHO classification, Arabic numerals are used, consistent with classification schemes for cancers in other organ systems. This change reduces the possibility of typographic error or misreading a pathology report, for example, when a grade ‘II’ tumour might be mistyped as, or mistaken for, a grade ‘III’ tumour¹. In the new classification scheme the term ‘anaplastic’ is no longer used; such tumours are instead simply referred to as grade 3 (REFS^{1,2}). It also bears emphasizing that, although the 2021 WHO classification still retains a scale of grades 1–4, most tumour types do not encompass all four grades — for example, none of the adult-type diffuse gliomas are WHO grade 1, and tumours such as glioblastoma and medulloblastoma are only WHO grade 4. Additionally, the term ‘CNS WHO grade’ has now been introduced to emphasize that these grading schemes are specifically coming from the editors of the CNS classification scheme and authors within the WHO, as opposed to the broader organization itself.

For ease of reading, however, the shorter ‘WHO grade’ or simply ‘grade’ is used throughout this Review.

Before 2016, the WHO classification schemes were predicated on the idea that the same criteria should be applied to CNS tumours worldwide. As access to advanced molecular testing was (and is) uneven across the world, this approach meant that the official scheme could incorporate only histological features, not molecular data. Over time such a position became untenable, as many key molecular alterations could not be reliably inferred through histopathological evaluation, even when performed by experts. For example, oligodendrogliomas were traditionally identified on the basis of round nuclei, hydropic changes in the cytoplasm, and delicate branching vasculature, whereas astrocytomas had more angulated nuclei with clumped chromatin. However, many tumours showed equivocal, or ‘hybrid’, histological features, leading to the ‘oligoastrocytoma’ entity. Subsequent research showed that oligoastrocytomas are actually either astrocytomas or oligodendrogliomas, but making reliable distinctions between the two requires screening for mutations in the genes encoding IDH1 and IDH2, as well as testing for chromosome 1p/19q codeletion, wherein oligodendrogliomas have IDH mutation and 1p/19q codeletion^{13,14}. Thus, the 2016 WHO classification was the first to incorporate IDH mutation and 1p/19q codeletion status into the official diagnostic terms for those tumours, retaining ‘oligoastrocytoma’ for rare tumours that still could not be placed into either entity⁴ (Supplementary Table 1). Classification as a glioblastoma required the presence of necrosis and/or microvascular proliferation, and was split into IDH-wildtype and IDH-mutant subcategories. A few other molecular alterations, involving *H3-3A*, *RELA*, *TP53* and *C19MC* were also included in the official diagnostic terms of specific tumours. In the 2021 CNS classification, this trend towards defining tumours on the basis of particular molecular alterations has expanded greatly, and molecular alterations in *MYB* or *MYBL1*, *MN1*, *YAP1*, *MYCN*, *FOXR2*, *BCOR*, *SMARCB1*, *FET-CREB* and *DICER1* are now included within specific tumour categories (BOX 1). Even among those tumours for which a specific alteration is not yet part of the official diagnostic term, well-recognized alterations are now known, the detection of which can greatly assist in the resolving of difficult cases (for example, the presence of *KIAA1549-BRAF* fusion in a pilocytic astrocytoma, or *PRKCA* mutation in a chordoid glioma).

Because access to the kind of testing that is required for many of these diagnoses is still lacking in large parts of the world, the 2021 WHO edition includes the terms ‘not otherwise specified’ (NOS) and ‘not elsewhere classified’ (NEC)^{1,2}. Although NOS and NEC might seem interchangeable, they refer to two very different situations. ‘NOS’ indicates that the molecular and/or immunohistochemical testing needed to precisely classify a particular CNS tumour by the new scheme is not available. For example, if a cerebellar tumour shows the histopathological features of a medulloblastoma, but there are no means by which to further subtype the tumour,

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Box 1 | Gliomas and embryonal tumours in the fifth edition of the WHO classification of CNS tumours

Gliomas

Adult-type diffuse gliomas

- Astrocytoma, IDH-mutant, WHO grades 2–4
- Oligodendroglioma, IDH-mutant and 1p/19q-codeleted, WHO grades 2–3
- Glioblastoma, IDH-wildtype, WHO grade 4

Paediatric-type diffuse gliomas

- Low-grade
 - Diffuse astrocytoma, *MYB*-altered or *MYBL1*-altered^a
 - Angiocentric glioma
 - Polymorphous low-grade neuroepithelial tumour of the young (PLNTY)^a
 - Diffuse low-grade glioma, MAPK pathway-altered^a
- High-grade
 - Diffuse midline glioma, H3 K27-altered
 - Diffuse hemispheric glioma, H3 G34-mutant^a
 - Diffuse paediatric-type high-grade glioma, H3-wildtype and IDH-wildtype^a
 - Infant-type hemispheric glioma^a

Circumscribed astrocytic gliomas

- Pilocytic astrocytoma
- High-grade astrocytoma with piloid features^a
- Pleomorphic xanthoastrocytoma, WHO grades 2–3
- Subependymal giant cell astrocytoma
- Chordoid glioma
- Astroblastoma, *MN1*-altered

Ependymal tumours

- Supratentorial ependymoma
 - *ZFTA*-fusion ependymoma
 - *YAP1*-fusion ependymoma^a
- Posterior fossa ependymoma
 - Posterior fossa ependymoma, group PFA^a
 - Posterior fossa ependymoma, group PFB^a
- Spinal ependymoma
- Spinal ependymoma, *MYCN*-amplified^a
- Myxopapillary ependymoma
- Subependymoma

Embryonal tumours

Medulloblastoma

- Medulloblastoma, WNT-activated
- Medulloblastoma, SHH-activated and *TP53*-wildtype
- Medulloblastoma, SHH-activated and *TP53*-mutant
- Medulloblastoma, non-WNT/non-SHH

Other

- Atypical teratoid–rhabdoid tumour
- Cribriform neuroepithelial tumour^a
- Embryonal tumour with multilayered rosettes
- CNS neuroblastoma, *FOXR2*-activated^a
- CNS tumour with *BCOR* internal tandem duplication^a
- CNS embryonal tumour

This is only a subset of all the tumour types and subtypes in the entire WHO classification scheme. ^aNew tumour types and subtypes.

‘medulloblastoma, CNS WHO grade 4, NOS’ can be used in the final diagnosis of the pathology report. By contrast, ‘NEC’ refers to cases in which advanced molecular testing was done, but still failed to classify the tumour. One such example would be a primitive, embryonal-appearing cerebral neoplasm that did not contain any diagnostic molecular alterations and did not map to a known cluster of tumours on DNA methylation profiling (FIG. 1). In that instance, the pathological diagnosis could be ‘embryonal tumour, NEC’. Another example would be a diffuse-appearing glioma that occurs in an adult, yet has neither the histopathological features of glioblastoma, nor any of the molecular features associated with either adult-type or paediatric-type diffuse gliomas, or even circumscribed astrocytic gliomas (discussed in more detail below). Such a case could be diagnosed as ‘diffuse glioma, IDH-wildtype, CNS WHO grade 2 or 3, NEC’.

Adult-type diffuse glioma

Pathological features of adult-type diffuse gliomas.

Diffusely infiltrating gliomas are the most common tumours to arise within the brain parenchyma of adults, accounting for more than 70% of all such neoplasms¹⁵.

Previous iterations of the WHO classification scheme broke these tumours down into grade 4 glioblastomas, grade 2–3 astrocytomas and grade 2–3 oligodendrogliomas. Until 2008, the most widely used molecular test in these tumours was to detect 1p/19q codeletion, the hallmark of oligodendroglioma¹⁶. Otherwise, tumour typing was based solely on histopathology, wherein astrocytomas had irregular angulated nuclei, and oligodendrogliomas had rounder nuclei with branched capillaries, hydropic change (‘fried egg’ cellular appearance), and mineralization. Astrocytic tumours with necrosis and/or microvascular proliferation were grade 4 glioblastomas, those with elevated mitotic activity but lacking necrosis and microvascular proliferation were grade 3 anaplastic astrocytomas, and tumours without necrosis, microvascular proliferation or elevated mitoses were grade 2. Oligodendrogliomas with necrosis, microvascular proliferation, and/or elevated mitoses were grade 3, and tumours lacking all three features were grade 2.

In 2008 and 2009, seminal papers were published that described point mutations in the gene encoding IDH1 and, to a lesser extent, the gene encoding IDH2. Tumours with mutations in either *IDH1* or *IDH2* are referred to as IDH-mutant, and tumours without such mutations are referred to as IDH-wildtype^{17,18}. In the years since those first descriptions of *IDH* mutations were published, other key aspects of IDH-mutant and IDH-wildtype gliomas have repeatedly been confirmed. First, we now know that IDH-mutant gliomas are fundamentally different from IDH-wildtype gliomas in terms of their metabolism, epigenetics, biological behaviour and response to therapy¹⁹. Second, diffusely infiltrative gliomas that are IDH-mutant are less aggressive than IDH-wildtype gliomas, even when an IDH-mutant astrocytoma has necrosis and/or microvascular proliferation, which are the traditional histological hallmarks of glioblastoma (the most aggressive glioma)²⁰. Third, IDH-mutant astrocytomas with homozygous deletion of the *CDKN2A* and/or *CDKN2B* genes — which are located next to each other on chromosome 9p21 and encode the cell cycle checkpoints p16 and p15, respectively — have a prognosis similar to that of IDH-mutant astrocytomas with necrosis and/or microvascular proliferation²¹. Fourth, aside from some rare cases, IDH mutations occur in adult patients, not paediatric patients, with the incidence increasing during adolescence²²; the average patient age is 38 years of age for IDH-mutant astrocytoma and 46 years of age for IDH-mutant oligodendroglioma (Supplementary Fig. 1). Although the incidence of IDH-mutant tumours decreases with advancing age, cases have been found in the elderly, even in octogenarians. Fifth, unlike the situation in adults, diffuse paediatric gliomas that are histologically lower grade often have *MYB* or *MYBL1* alterations, as opposed to IDH mutations²³. Sixth, true whole-arm 1p/19q codeletion of the type associated with oligodendrogliomas, arising from unbalanced translocation, is always seen in conjunction with an IDH mutation^{24,25} (Supplementary Fig. 2). Seventh, within brain tumours, IDH mutations are found only in diffusely infiltrative glioma (either diffuse astrocytoma or oligodendroglioma). Astrocytic tumours that might recur locally but do not migrate on

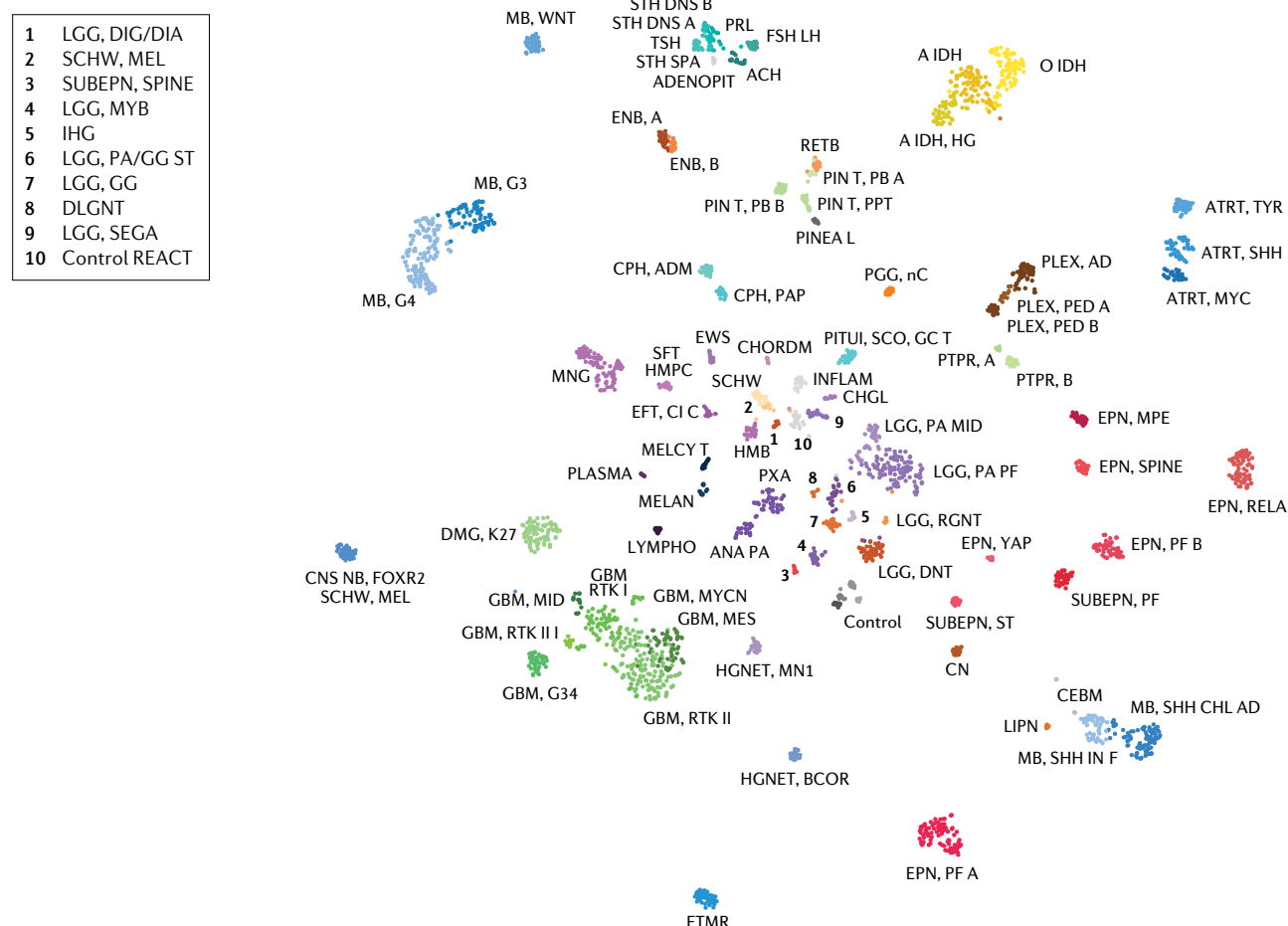
a Reference cohort (91 classes)

Embryonal 1 ETMR 1 MB, WNT 2 MB, G3 2 MB, G4 2 MB, SHH CHL AD 2 MB, SHH INF 2 ATRT, MYC 2 ATRT, SHH 2 ATRT, TYR 2 CNS NB, FOXR2 5 HGNET, BCOR	Glioblastoma 1 DMG, K27 2 GBM, G34 2 GBM, MES 2 GBM, RTK I 2 GBM, RTK II 2 GBM, RTK III 2 GBM, MID 2 GBM, MYCN	Glioneuronal 1 CN 1 DLGNT 1 LIPN 1 LGG, DIG/DIA 1 LGG, DNT 1 LGG, RGNT 1 RETB 2 ENB, A 2 ENB, B 2 PGG, nC 4 LGG, GG	Sella 1 CPH, ADM 1 CPH, PAP 1 PITAD, ACH 1 PITAD, FSH LH 1 PITAD, PRL 1 PITAD, STH SPA 1 PITAD, TSH 2 PITAD, STH DNS A 2 PITAD, STH DNS B 4 PITUI, SCO, GCT	Other glioma 1 CHGL 1 LGG, SEGA 2 LGG, PA PF 2 LGG, PA MID 5 ANA PA 5 HGNET, MN1 5 IHG 4 LGG, MYB 4 LGG, PA/GG ST 4 PXA	Ependymal 1 EPN, RELA 2 EPN, YAP 2 EPN, PF A 2 EPN, PF B 2 EPN, SPINE 4 EPN, MPE 4 SUBEPN, PF 4 SUBEPN, SPINE 4 SUBEPN, ST
Nerve 1 SCHW 1 SCHW, MEL	Melanocytic 1 MELAN 1 MELCYT	Glioma IDH 3 A IDH 3 A IDH, HG 3 O ID	Mesenchymal 1 CHORDM 1 EWS 1 HMB 4 MNG 3 SFT HMPC 5 EFT, CIC	Pineal 2 PTPR, A 2 PTPR, B 2 PIN T, PB B 4 PIN T, PB A 3 PIN T, PPT	Control ADENOPIT WM CEBM HEMI HYPTHAL INFLAM PINEAL PONS REACT
Haematopoietic 1 LYMPHO 1 PLASMA	Plexus 3 PLEX, AD 3 PLEX, PED A 3 PLEX, PED B				

Relation to WHO entities (category):

1 Equivalent 2 Subclass 3 Not equivalent (combining grades) 4 Not equivalent (combining entities) 5 Not recognized by WHO

b t-SNE dimensionality reduction (2,801 samples)



a single-cell basis throughout the entire brain (for example, pilocytic astrocytomas, pleomorphic xanthoastrocytomas, chordoid gliomas and astroblastomas) do not have IDH mutations^{1,26}. Last, IDH-wildtype diffuse gliomas in adults usually contain activating alterations in genes encoding receptor tyrosine kinases, such as *EGFR*,

as well as *TERT* promoter mutations and concomitant gain of chromosome 7 and loss of chromosome 10. Such alterations strongly suggest a glioblastoma prognosis, even if the tumour does not show the histopathological features of glioblastoma^{1,2}. Conversely, IDH-wildtype diffuse high-grade gliomas driven by mutations in

Fig. 1 | Methylation patterns of CNS tumours. **a** | Key to the t-distributed stochastic neighbour embedding (t-SNE) plot shown in part **b**. Abbreviations for each tumour type can be found in the supplementary section of REF.¹²⁹. Major tumour categories are sorted by colour as follows: embryonal tumours, shades of blue; IDH-wildtype glioblastomas and H3-altered gliomas, shades of green; other (usually MAPK-activated) gliomas, shades of violet; ependymomas, shades of red; glioneuronal tumours, shades of orange; IDH-mutated gliomas, shades of yellow; choroid plexus tumours, shades of brown; pineal tumours, shades of mint green; melanocytic tumours, shades of dark blue; sellar region tumours, shades of cyan; mesenchymal tumours, shades of pink; nerve tumours, shades of beige; haematopoietic tumours, shades of dark purple; control tissues: shades of grey. **b** | Genomic DNA methylation pattern analyses reveal that CNS tumours of each type cluster together, and apart from other tumour types. In this t-SNE plot, which is a two-dimensional depiction of multidimensional methylation data, each data point refers to an individual tumour. Figure adapted from REF.¹²⁹, Springer Nature Ltd.

H3-3A (formerly *H3F3A*) are far more common in paediatric patients than in adult patients with glioma^{27–30}.

The 2016 WHO classification had all the aforementioned tumours listed under ‘Diffuse astrocytic and oligodendroglial tumours’, ‘Other astrocytic tumours’ and ‘Other gliomas’ without any indication of patient age or likely tumour behaviour⁴. However, the 2021 WHO classification uses ‘Adult-type diffuse gliomas’, ‘Paediatric-type diffuse low-grade gliomas’, ‘Paediatric-type diffuse high-grade gliomas’, and ‘Circumscribed astrocytic gliomas’ (BOX 1 and Supplementary Table 1). Only three subsets of adult-type diffuse glioma are now recognized, including ‘Astrocytoma, IDH-mutant’, ‘Oligodendroglioma, IDH-mutant and 1p/19q-codeleted’, and ‘Glioblastoma, IDH-wildtype’. Thus, a glioma that is IDH-mutant and 1p/19q-codeleted would be called ‘Oligodendroglioma, IDH-mutant and 1p/19q-codeleted’ and would be considered an adult-type diffuse glioma, even if it occurred in a child. Conversely, a high-grade glioma that has an *H3-3A* mutation in an adult would still be grouped under ‘Paediatric-type diffuse high-grade gliomas’.

Among adult-type diffuse gliomas, the most notable change in the 2021 edition is the reclassification of ‘Glioblastoma, IDH-mutant, WHO grade 4’ as ‘Astrocytoma, IDH-mutant, WHO grade 4’. This change was made to emphasize that IDH-mutant astrocytomas have a fundamentally different biology and less aggressive disease course than IDH-wildtype diffuse astrocytomas, even when the former contain necrosis and/or microvascular proliferation^{20,31}. This new category includes all IDH-mutant astrocytomas with homozygous *CDKN2A/B* deletion, whether they have grade 4 histopathological features or not³². Conversely, ‘glioblastoma’ should now refer only to diffusely infiltrative gliomas that are IDH-wildtype and have either the classic histopathological findings of necrosis and/or microvascular proliferation, or one or more of the following molecular alterations: *TERT* promoter mutation, *EGFR* amplification, concomitant gain of chromosome 7 and loss of chromosome 10 (or just 10q) (Supplementary Fig. 2)^{1,2}. Thus, an IDH-wildtype glioma that shows only grade 3 histological features (mitoses, but no necrosis or microvascular proliferation), or even one that has rare or no visible mitoses, can be upgraded to a glioblastoma through advanced molecular testing^{1,2} (FIG. 2). In a busy neuropathology practice, this happens on a fairly regular

basis — enough to warrant routine screening of all gliomas for these, and other, molecular markers. Thus, more pathology laboratories must be able to perform such tests, and third-party payers need to cover such costs³³.

Clinical aspects of adult-type diffuse gliomas. The new classification of adult-type diffuse gliomas should reduce the heterogeneity in prognosis and treatment responses among these tumours³⁴, should improve the reliability of outcomes in new clinical trials, and should provide a deeper understanding of the subgroups of patients who have a greater response to therapy³⁵. For example, in a phase III trial (RTOG 9802) that compared isolated radiation therapy with radiation therapy plus a combination of procarbazine, lomustine and vincristine (known as PCV), the addition of PCV chemotherapy extended overall survival only in patients with IDH-mutant gliomas, not in patients with IDH-wildtype glioblastomas³⁵. Nevertheless, an increased reliance on tumour genotype for accurate classification inevitably leads to longer delays in reaching a diagnosis and initiating either standard therapy or enrolment into clinical trials. Therefore, neuropathology laboratories need to have ready access to the appropriate technology, and must implement workflows that allow tumours to be processed and tested expeditiously. As the efficacy of targeted therapies improve, and patients have more treatment options, the need for rapid molecular testing will be even greater.

Although the term ‘glioblastoma’ now includes IDH-wildtype astrocytomas with molecular but not histopathological features of glioblastoma, it is not yet clear whether such tumours have exactly the same overall biology and response to treatment as IDH-wildtype gliomas with overt necrosis and/or microvascular proliferation. Findings published in the last 3 years suggest that, compared with histologically defined glioblastomas, molecularly defined glioblastomas might be more infiltrative and show less contrast enhancement on MRI, and might respond less well to temozolomide chemotherapy^{35–37}. Thus, although the current recommendation for glioblastoma is still radiation therapy with concomitant and adjuvant temozolomide^{38,39} (Supplementary Table 2), continued study of molecularly defined glioblastomas is needed to determine whether temozolomide chemotherapy is the best treatment option and whether glioblastomas defined in this way should be routinely grouped with histologically defined glioblastomas in clinical trials.

Standard treatment for IDH-wildtype glioblastoma is maximal safe resection followed by radiation therapy (60 Gy over 6 weeks) and daily concomitant temozolomide³⁸. This initial treatment is followed by 6 months of adjuvant temozolomide and, in some countries, electromagnetic tumour treating fields⁴⁰ (FIG. 3). The current recommendation for grade 4 IDH-mutant astrocytoma is that they should be treated with radiation therapy and temozolomide, similar to IDH-wildtype glioblastoma⁴¹ (FIG. 4). As patients with grade 4 IDH-mutant astrocytoma have better outcomes than those with IDH-wildtype astrocytoma³¹, and are now being excluded from new IDH-wildtype glioblastoma trials, it will be important to develop trials that specifically study IDH-mutant grade 4 astrocytoma.

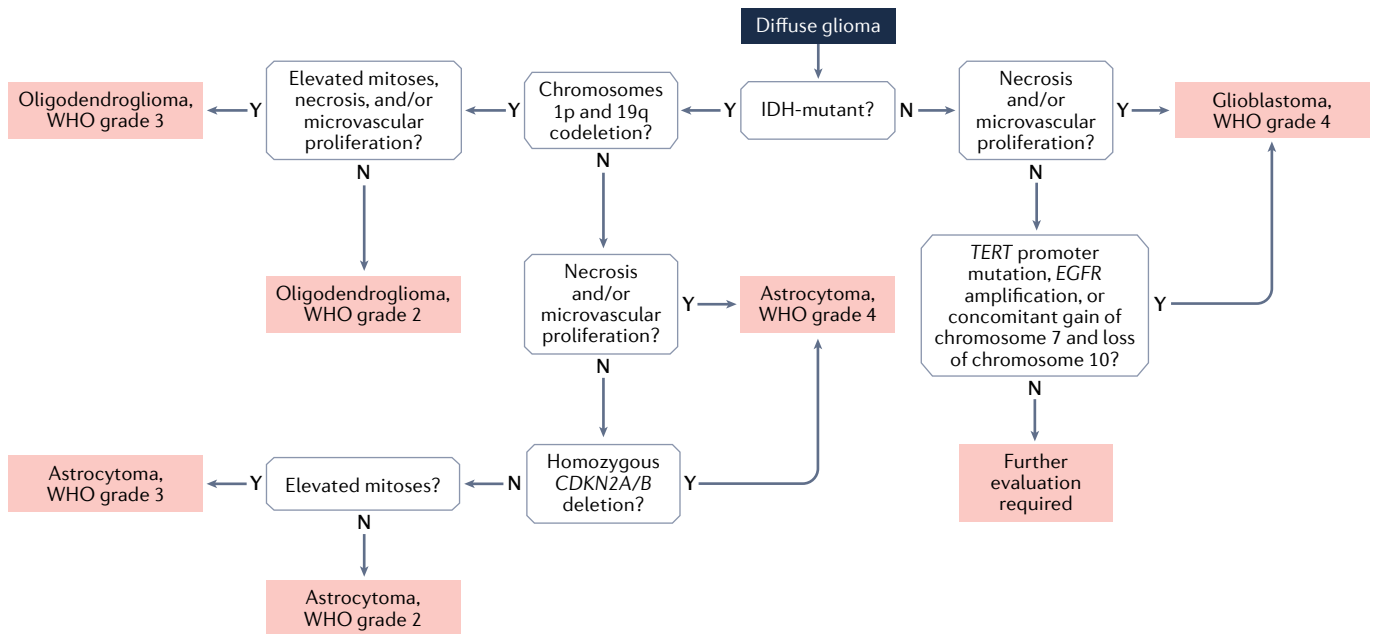


Fig. 2 | Decision tree for the evaluation of adult-type diffuse gliomas. The entry point of this decision tree is the identification of a diffuse glioma in an adult (black). The first step is to establish whether the tumour has mutations in isocitrate dehydrogenase (IDH) genes. Gliomas with IDH mutations are then stratified according to chromosomes 1p and 19q codeletion status. Tumours with whole-arm 1p/19q codeletion are oligodendrogliomas and are either WHO grade 2 or WHO grade 3 depending on the presence or absence of high-grade histological features, including increased mitoses, necrosis and/or microvascular proliferation. IDH-mutant gliomas without 1p/19q codeletion are astrocytomas, and are WHO grade 4 (necrosis, microvascular proliferation and/or homozygous

CDKN2A/B deletion), WHO grade 3 (only elevated mitoses) or WHO grade 2 (none of these features). IDH-wildtype gliomas with necrosis and/or microvascular proliferation are WHO grade 4 glioblastomas. IDH-wildtype gliomas without these histological features can still be diagnosed as WHO grade 4 glioblastomas if they have *TERT* promoter mutation, *EGFR* amplification, and/or concomitant gain of chromosome 7 and loss of chromosome 10 (or just 10q). If an IDH-wildtype glioma has neither the histological nor the molecular features of a glioblastoma, consideration of other tumours (for example, paediatric-type diffuse gliomas, circumscribed astrocytic gliomas, glioneuronal tumours, ependymomas) might be warranted. N, no; Y, yes.

For grade 2–3 IDH-mutant astrocytomas and oligodendrogliomas, the new classification should reduce the variation in patient outcomes within each glioma type³⁴. This increased consistency raises the possibility of treating all grade 2–3 (and perhaps even grade 4) IDH-mutant astrocytomas in the same way, and enrolling patients with any grade of IDH-mutant astrocytoma into the same clinical trials; the same possibility applies to grades 2–3 IDH-mutant and 1p/19q-codeleted oligodendrogliomas. For IDH-mutant astrocytomas, the current treatment recommendation is radiation therapy followed by 1 year of adjuvant temozolomide^{36,42} (FIG. 4 and Supplementary Table 2), whereas the value of concomitant temozolomide during radiation therapy remains a matter of debate³⁶. For both grade 2 and grade 3 oligodendrogliomas, radiation therapy followed by up to six cycles of PCV is recommended^{43–46}, although temozolomide is also an option and is generally better tolerated^{38,46}. Direct comparison of the efficacy of PCV versus temozolomide in grades 2 and 3 oligodendrogliomas is currently underway in the CODEL trial⁴⁷.

Paediatric-type diffuse gliomas

Pathological features of paediatric-type diffuse gliomas. The tripartite scheme for subtyping adult-type diffuse gliomas (IDH-mutant, IDH-mutant with 1p/19q codeletion, IDH-wildtype) had already been established in the 2016 edition of the WHO classification⁴. Aside from

the adjustments discussed above, little else has changed regarding those tumours in the 2021 edition. In contrast, the classification of paediatric gliomas has undergone substantial revision, reflecting the growth in our understanding of that subject. Paediatric gliomas have now been moved to a separate category of diffuse glial tumours, distinct from adult-type gliomas, and have been further subdivided into low-grade and high-grade groups. This reclassification represents a continuation of the emphasis on molecular features that began in the 2016 edition, in which ‘diffuse intrinsic pontine glioma’ became ‘diffuse midline glioma, H3 K27M-mutant’⁴. In the 2021 edition, that tumour — now called ‘diffuse midline glioma, H3 K27-altered’ to include tumours lacking H3 mutations but still showing loss of H3 K27 me3 immunostaining — is accompanied by three other types of paediatric high-grade glioma: ‘diffuse hemispheric glioma, H3 G34-mutant’, ‘diffuse paediatric-type high-grade glioma, H3-wildtype and IDH-wildtype’ and ‘infant-type hemispheric glioma’ (BOX 1).

Three new tumours have also been added among the low-grade paediatric gliomas, including ‘diffuse astrocytoma, *MYB*-altered or *MYBL1*-altered’, ‘polymorphous low-grade neuroepithelial tumour of the young’ (PLNTY), and ‘diffuse low-grade glioma, MAPK pathway-altered’. All three of these tumours can show single-cell integration into the brain parenchyma; hence they are termed ‘diffuse’ gliomas. However, unlike

adult-type diffuse gliomas or paediatric-type diffuse high-grade gliomas, many patients with lower-grade paediatric-type diffuse gliomas demonstrate indefinite recurrence-free survival after resection. In this context, then, the term 'diffuse' is perhaps best understood as 'that which does not have a discrete tumour–non-tumour border' as opposed to 'that which clearly and insidiously infiltrates throughout the brain, beyond what can be surgically resected'. Distinguishing between these two connotations of 'diffuse' via traditional light microscopy can be very difficult, and hence the need for advanced molecular diagnostics, integrated with thorough radiological evaluation.

Even when incorporating molecular data, the lines separating the four paediatric-type low-grade diffuse gliomas are conceptually blurry. Although they do segregate into fairly distinct methylation clusters, they share similar kinds of driver mutations, histopathological features, and clinical outcomes. For example, 'diffuse astrocytoma, *MYB*-altered or *MYBL1*-altered' is necessarily defined as having *MYB* or *MYBL1* alterations, yet angiocentric gliomas also usually have *MYB* fusions. In both types of tumour, these alterations

result in robust MAPK pathway activation. Likewise, the polymorphous PLNTY neoplasm contains histological elements of several other tumours, including oligodendrogliomas, gangliogliomas, pilocytic astrocytomas, dysembryoplastic neuroepithelial tumours and pleomorphic xanthoastrocytomas. Similar to most of those other tumours, PLNTY also features MAPK-activating mutations, this time in the form of *BRAF*, *FGFR2*, *FGFR3* or even *NTRK* alterations. 'Diffuse low-grade glioma, MAPK pathway-altered' looks very similar to the other aforementioned tumours, and its growth is mainly driven by *BRAF* mutations or *FGFR1* alterations (although other MAPK-activating driver mutations have also been found⁴⁸). Whether all four of these diagnoses will remain distinct in future iterations of the WHO classification scheme is unclear, but they are all united by MAPK pathway activation, as well as by a lack of IDH mutations or mutations in histone-encoding genes. Furthermore, although all four tumour types are low-grade, accompanying alterations of cell cycle-associated genes, such as *CDKN2A/B* deletion, suggest faster recurrence and greater risk of transformation to higher-grade behaviour^{1,2}.

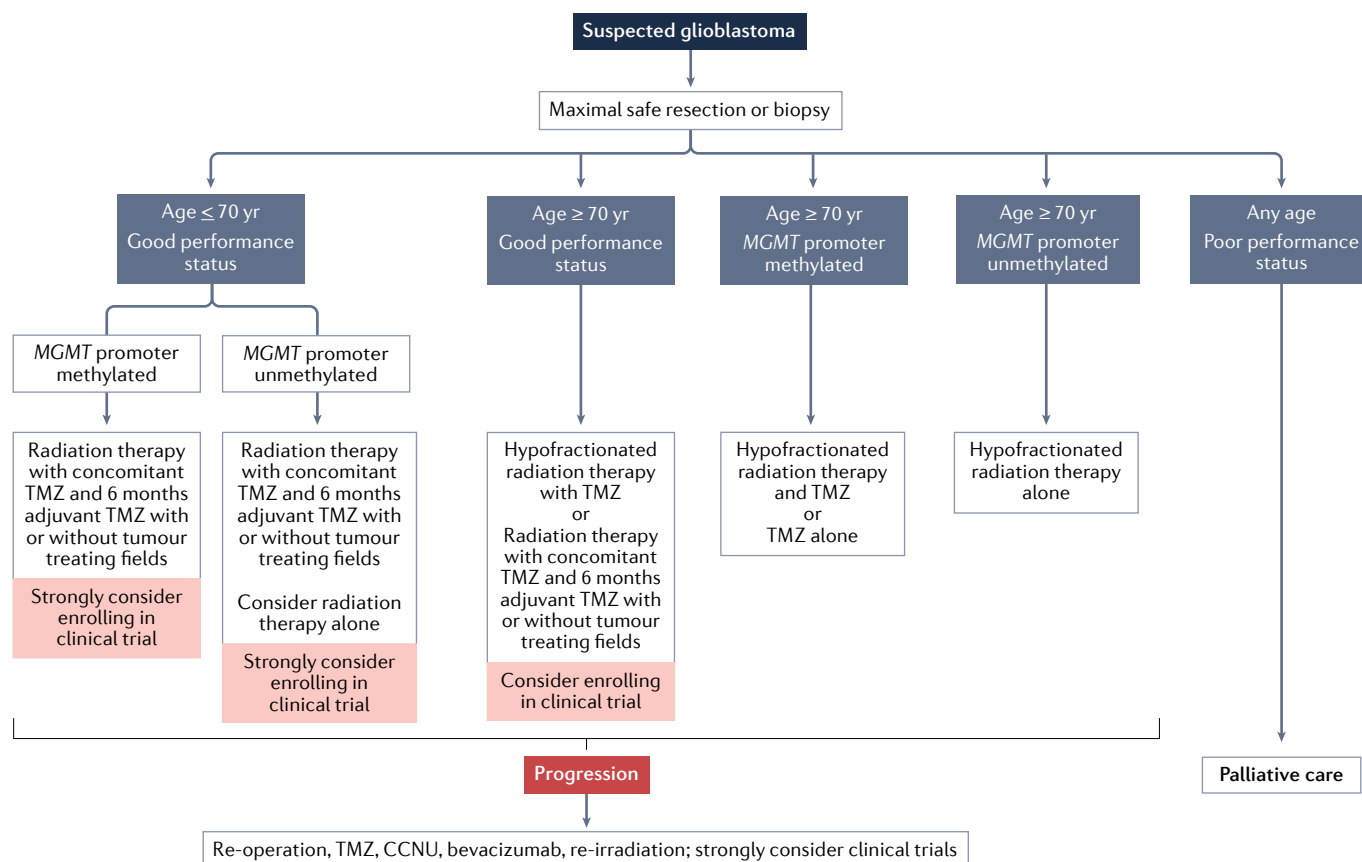


Fig. 3 | Therapy for glioblastoma, IDH-wildtype, WHO grade 4. Flow chart showing suggested therapeutic approaches for IDH-wildtype glioblastoma, WHO grade 4. Treatment options for patients with glioblastoma largely depend on three variables: patient age, overall functionality (as indicated by Karnofsky performance status), and MGMT promoter methylation status of the tumour. Younger and/or more functional patients generally have more treatment options, including radiation, temozolomide (TMZ), electromagnetic tumour treating fields, and clinical trials. Patients ≥ 70 years

old whose glioblastoma lacks MGMT promoter methylation should be offered only radiotherapy, and only palliative care is recommended for low-functioning patients of any age. After front-line therapy, individuals with recurrent glioblastoma should ideally be offered enrolment into clinical trials. Other options include re-resection, more TMZ, more radiation, lomustine and/or bevacizumab. Radiation therapy consists of 60 Gy in 30 fractions; hypofractionated radiation therapy consists of 40 Gy in 15 fractions. CCNU, lomustine; MGMT, O⁶-methylguanine-DNA methyltransferase.

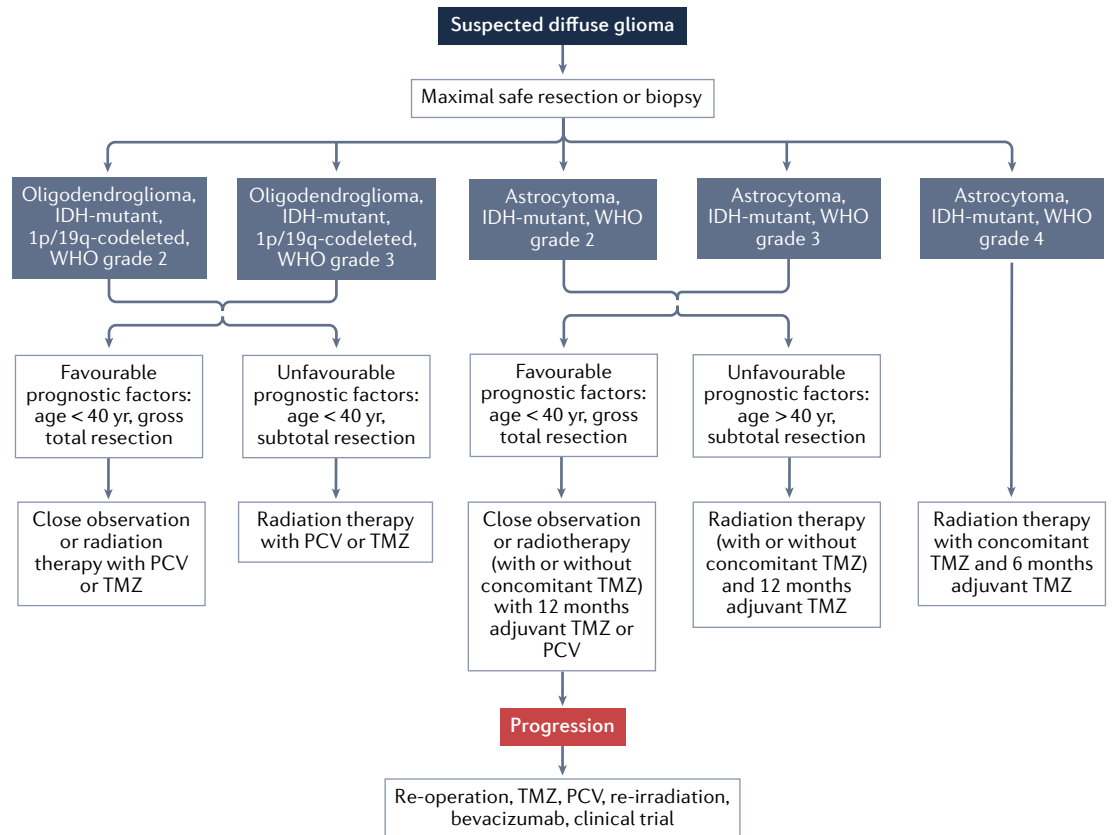


Fig. 4 | **Therapy for IDH-mutant glioma.** Flow chart showing suggested therapeutic approaches for isocitrate dehydrogenase (IDH)-mutant glioma. Patients <40 years old with grade 2 IDH-mutant, chromosomes 1p and 19q codeleted oligodendroglioma with gross total resection of their tumour can simply be observed for recurrence. Patients <40 years old with grade 3 IDH-mutant oligodendroglioma with gross total resection are generally given radiation plus PCV (procarbazine, lomustine and vincristine) or temozolomide (TMZ) chemotherapy but occasionally can be observed. In patients >40 years old and/or whose tumours could not be extensively resected, radiation with PCV or TMZ is recommended. Patients with WHO grade 2–3 IDH-mutant astrocytoma can be managed in a similar manner to those with oligodendroglioma based on patient age and extent of tumour resection, although TMZ rather than PCV is preferred. If an IDH-mutant astrocytoma is grade 4, radiation plus TMZ is recommended regardless of the extent of neurosurgical resection. After front line therapy, recurrent IDH-mutant astrocytomas can be treated with re-resection and more TMZ and radiation, PCV, or bevacizumab (clinical trials might also be considered).

Among high-grade paediatric-type diffuse gliomas, ‘diffuse hemispheric glioma, H3 G34-mutant’ is essentially the hemispheric counterpart to the midline glioma driven by H3 K27 alterations (Supplementary Fig. 3). For reasons that remain ambiguous, K27 mutations preferentially arise in midline gliomas, whereas G34 mutations are present in hemispheric tumours³⁰. Both mutations cause a loss of histone trimethylation, but G34-mutant hemispheric gliomas often have *MGMT* promoter methylation, whereas K27-mutant midline gliomas do not. Prognosis is poor in both types, especially midline gliomas. ‘Diffuse paediatric-type high-grade glioma, H3-wildtype and IDH-wildtype’ is a somewhat awkward term reflecting the fact that some diffuse high-grade gliomas, preferentially arising in children, lack histone mutations or IDH mutations. These tumours can look like glioblastomas or undifferentiated, primitive-appearing tumours (Supplementary Fig. 4a,b). Their driver mutations do resemble those of IDH-wildtype glioblastomas, involving *PDGFRA*, *EGFR* or *NF1* (among others), but their genomic DNA methylation patterns are distinct

from those of IDH-wildtype glioblastomas. Finally, there are the ‘infant-type hemispheric gliomas’, which are different from desmoplastic infantile ganglioglioma or desmoplastic infantile astrocytoma (Supplementary Fig. 4c–f). Most infant-type hemispheric gliomas seem to have isolated receptor tyrosine kinase-activating fusions, rendering them highly sensitive to targeted inhibition of whichever receptor tyrosine kinase is altered^{49–51}. These tumours are considered high-grade on the basis of their appearance and a reported 5-year overall survival of 25–50%^{1,2}.

Clinical aspects of paediatric-type diffuse gliomas. Until 2021, ‘diffuse low-grade glioma, MAPK pathway-altered’, which comprises a minority of all paediatric low-grade gliomas, was included in both retrospective and prospective studies of paediatric low-grade gliomas; this lack of separation has limited progress in their management^{52,53} (TABLE 1). Whether these tumours, which are often non-enhancing and ill-defined on MRI, respond to conventional chemotherapy (for example,

carboplatin and vincristine or vinblastine) as well as the more circumscribed (usually pilocytic) astrocytomas do is unknown. Given that many tumours defined as 'diffuse low-grade glioma, MAPK pathway-altered' have *BRAF* mutations^{48,54}, other targeted therapeutic options are available. Prospective randomized trials comparing the disease control and functional outcomes achieved with standard chemotherapy with those achieved by dual molecular targeted therapy (for example, *BRAF* inhibitor and MEK inhibitor combinations)^{55–57}, or MEK inhibitors as monotherapy^{58–60}, could dramatically alter management of these tumours. However, until completion of randomized trials, chemotherapy with carboplatin and vincristine or, in selected cases, focal radiation (especially in older children with relatively localized tumours), remains standard of care³⁹. Second generation *BRAF* inhibitors for treatment of these tumours are already in clinical trials, as are *FGFR* inhibitors^{61–63}. The other paediatric-type diffuse low-grade gliomas remain clinically problematic, as they are even less common. Aside from 'watchful waiting' when the tumour is low-grade and has gross total or near-total resection, the optimal adjuvant treatments for those tumours, including specific chemotherapies and/or radiotherapy regimens, remain unclear.

Among paediatric-type diffuse high-grade gliomas, H3 G34-mutant tumours still have a poor prognosis even if the *MGMT* promoter is methylated (which predicts

response to alkylating agents such as temozolomide)⁶⁴; therapeutic targeting of the other molecular abnormalities that are found in these tumours, such as *PDGFR* mutations, is currently being explored⁶⁵. Although it produces suboptimal outcomes, the conventional therapy for these tumours remains focal radiation and post-radiation alkylating drugs⁶⁶. The data to support use of temozolomide in paediatric patients during radiation therapy is equivocal⁶⁷. Entry into an Institutional Review Board-approved investigational trial is optimal, but when such trials are not available, evidence from one study indicates that treatment with temozolomide during radiotherapy, followed by dual alkylator therapy with temozolomide and lomustine, might provide a survival advantage⁶⁶. Probably the most important therapeutic change resulting from the new 2021 WHO classification of paediatric high-grade gliomas is related to the recognition of infant-type hemispheric gliomas as a distinct tumour type. The inclusion of this new tumour underscores the clinical impression that a subgroup of very young children with paediatric high-grade gliomas achieve surprisingly good outcomes. Because infant-type hemispheric gliomas have distinct molecular features, they require different therapeutic approaches than other paediatric-type diffuse high-grade gliomas. Indeed, preliminary evidence indicates that at least some infant-type hemispheric gliomas with *NTRK* fusions respond well to selective molecular inhibitors⁶⁸.

Table 1 | Recommended treatment for paediatric-type diffuse gliomas

Paediatric diffuse gliomas	WHO grade	Molecular alterations	Post-resection treatment recommendations	
			Total or near total resection	Subtotal resection
Low-grade gliomas				
Diffuse astrocytoma (MYB-altered or MYBL1-altered)	1	MYB, MYBL1	Observation	Carboplatin and vincristine ^a or vinblastine, possible focal radiotherapy
Angiocentric glioma	1	MYB	Observation	Carboplatin and vincristine ^a or vinblastine, possible focal radiotherapy
Polymorphous low-grade neuroepithelial tumour of the young	1	BRAF, FGFR	Observation	Consideration for BRAF/MEK inhibitor combination therapy or FGFR inhibitors, possible focal radiotherapy ^a
Diffuse low-grade glioma (MAPK pathway-altered) ^b	NA (low grade)	MAPK pathway	Observation	Carboplatin and vincristine ^a or vinblastine, possible molecular targeted therapy, possible focal radiotherapy
High-grade gliomas				
Diffuse midline glioma, H3 K27-altered	4	H3 K27, EGFR, EZHIP	Focal radiation therapy 54–59 Gy, and investigational drugs, if available	
Diffuse hemispheric glioma, H3 G34-mutant	4	H3 G34	Focal radiation therapy 54–59 Gy, and temozolomide ± lomustine, or investigational therapy	
Diffuse paediatric-type high-grade glioma, H3-wildtype, IDH-wildtype	4	IDH-wildtype, H3-wildtype, EGFR, PDGFRA, MYCN, methylome	Focal radiation therapy 54–59 Gy with or without temozolomide chemotherapy or investigational therapy ± lomustine, or other adjuvant therapy ^a	
Infant-type hemispheric glioma	NA (high grade)	RTK gene fusions	Individualized: potential targeted therapy ^b	

NA, not applicable; ^aGiven the risk of long-lasting polyneuropathies, no single dose of vincristine should be higher than 2 mg, and therapy is usually stopped after a cumulative dose of 8–12 mg. ^bConsideration of clinical trial, if available.

Circumscribed astrocytic gliomas

Pathological features of circumscribed astrocytic gliomas. An important conceptual advance in the new 2021 WHO classification is the discrimination between gliomas with indistinct tumour borders and gliomas in which the tumour–non-tumour border is comparatively sharper. Five well-established tumours are now assigned to the latter group, including pilocytic astrocytoma, pleomorphic xanthoastrocytoma, subependymal giant cell astrocytoma, chordoid glioma, and astroblastoma, *MNI*-altered (Supplementary Fig. 5a–f). The only new tumour in this group is ‘high-grade astrocytoma with piloid features’, in which the astrocytic neoplasm has MAPK-activating alterations (such as *NF1* inactivation, *BRAF* fusions or activating *FGFR1* mutations), homozygous loss of *CDKN2A/B*, and, usually, *ATRX* inactivation (Supplementary Fig. 5g,h); however, the most definitive way to diagnose this tumour is by methylation profiling⁶⁹. Previously, many of these tumours would have been called ‘anaplastic pilocytic astrocytomas’, but genomic DNA methylation profiling showed that high-grade astrocytomas with piloid features form a distinct methylation cluster⁶⁹. Necrosis and mitoses are often found in these tumours, and the 5-year overall survival rate is about 50%⁶⁹, hence the ‘WHO grade 3’ designation.

Many of the circumscribed astrocytic gliomas share a number of molecular and morphological features with paediatric-type diffuse low-grade gliomas (for example, MAPK-activating alterations and mineralization), to the point where a future edition of the WHO classification scheme might combine some or most of them into an entirely new group on the basis of commonly altered signalling pathways, DNA methylation cluster proximity, and/or biological behaviour.

Clinical aspects of circumscribed astrocytic gliomas.

Circumscribed astrocytic gliomas are relatively common in children, but can also occur in young adults¹. Compared with paediatric-type diffuse high-grade gliomas, circumscribed astrocytic gliomas have more established therapies and better prognoses. This is especially the case for pilocytic astrocytomas, which are predominantly driven by *BRAF* fusions. Clinical studies have demonstrated often dramatic responses of *BRAF* fusion pilocytic astrocytomas to MEK inhibitors^{53,70}. An additional trial comparing conventional therapy (carboplatin and vincristine) to these MEK inhibitors in children with progressive newly diagnosed *BRAF* fusion or *NF1*-mutant pilocytic astrocytomas is now underway⁷¹. Until these studies are completed, conventional chemotherapy with carboplatin and vincristine in infants and young children, and possibly vinblastine in older patients, remains the standard of care^{72,73}. Less frequently, focal radiation is used to treat sporadic pilocytic astrocytomas, especially when tumours are located in critical areas of the brain such as the brainstem⁵². Radiation therapy is used infrequently in children with *NF1*-related pilocytic astrocytomas because of concerns about radiation-induced mutagenesis^{39,40}.

For other circumscribed astrocytic gliomas (or even glioneuronal tumours) — such as pleomorphic

xanthoastrocytoma, where *BRAF*^{V600E} occurs in 60–80% of cases⁷⁴ — *RAF* inhibitors such as vemurafenib⁷⁵, and *RAF* and *MEK* inhibitor combinations such as dabrafenib and trametinib, can produce durable response rates approaching 70%⁷⁶. Unlike the situation in children with pilocytic astrocytomas, the data for the efficacy of any form of chemotherapy or radiotherapy in children with other circumscribed astrocytic gliomas is far from conclusive⁷⁷. In these children, immediate use of molecular-targeted therapy might be optimal³⁹. The introduction of the category ‘high-grade astrocytoma with piloid features’ is another step forward, as these tumours occur in patients with either *BRAF* abnormalities or *NF1* inactivation, although neither *ATRX* alterations nor *CDKN2A/B* deletion are yet directly targetable with therapy. Nevertheless, if any progress is to be made in managing these tumours, they must be identified and treated differently from low-grade gliomas without *CDKN2A/B* deletions or *ATRX* alterations.

Ependymal tumours

Pathological features of ependymal tumours. Similar to paediatric-type gliomas, the classification of ependymomas has undergone substantial revisions in the new edition of the WHO classification scheme. These revisions are predicated on molecular alterations that confer markedly different prognoses^{1,7} (BOX 1), and such alterations are strongly associated with the site at which the ependymoma arises^{78,79}. For example, many supratentorial ependymomas will have a fusion involving either *ZFTA* or *YAPI*, both of which are helpful in discriminating ependymomas from morphologically similar neoplasms driven by *FOXR2*, *BCOR* or *MNI* alterations^{1,2}. Although the 2016 WHO classification scheme referred to *RELA* fusion-positive ependymomas, the *ZFTA* gene is the main gene involved in this fusion (with *RELA* as the most common fusion partner for *ZFTA*)⁸⁰, so this tumour has been renamed ‘supratentorial ependymoma, *ZFTA* fusion-positive’ in the 2021 edition. Ependymomas with *ZFTA* fusions are more common, and seem to be more aggressive, than those with *YAPI* fusions⁸⁰.

Posterior fossa ependymomas are split into two groups, posterior fossa A (PFA) and posterior fossa B (PFB), on the basis of different DNA methylation patterns. Compared to PFA ependymomas, PFB ependymomas have more H3 K27 trimethylation, more whole-chromosome abnormalities, and a better prognosis⁸¹. Both spinal ependymomas and PFB ependymomas have frequent chromosome 22q (*NF2*) loss, but spinal ependymomas have a DNA methylation cluster distinct from either PFA or PFB tumours. Most spinal ependymomas look and act even more benign than PFB ependymomas, although a rare subset with *MYCN* amplification is histologically high-grade and aggressive, spreading quickly throughout the CSF⁸².

Particularly noteworthy in the new scheme are myxopapillary ependymomas. For years they were listed as WHO grade 1 and were often thought to be spinal ependymomas that were undergoing senescence as indicated by their vascular sclerosis, myxoid background, and degenerative-type nuclear atypia. However, these

tumours have their own methylation pattern, distinct from that of other spinal ependymomas, and can locally recur and sometimes even disseminate, especially in children^{83,84}. Thus, in the new edition of the WHO scheme, myxopapillary ependymomas are now listed as grade 2. Subependymomas, in contrast, remain WHO grade 1.

Clinical aspects of ependymal tumours. The separation of PFA from PFB ependymomas has important therapeutic connotations. Both PFA and PFB tumours can be difficult to remove, and attempts at complete surgical resection might cause considerable morbidity. Compared with PFA tumours, PFB tumours have a better prognosis and are more likely to be amenable to treatment, even after first relapse, which has surgical implications⁸⁵ (TABLE 2). One difficulty facing the appropriate diagnosis of these tumours is that they are best separated on the basis of genomic DNA methylation patterns, but such testing is not yet ubiquitous. However, H3K27me3 immunostaining is much lower in PFA ependymomas than in PFB ependymomas, which might help to discriminate between the two tumour types when methylation testing is not available⁸⁶. Negative prognostic markers not yet incorporated into the WHO scheme, such as *TERT* promoter mutations and chromosome 1q gain or 6q loss, are also being used to guide therapy in PFA and PFB ependymomas^{87–89}. Standard therapy for all posterior fossa ependymomas is gross total resection (if feasible) followed by focal radiation therapy⁵³. Proton beam radiation has been used to treat these tumours; evidence indicates that its efficacy is at least equal to that of focal radiation therapy⁹⁰, and it also has the

advantage of exposing the brain and surrounding structures to less ‘scatter’ radiation (similar treatment might also be considered for circumscribed astrocytomas and/or low-grade paediatric-type diffuse gliomas). Although chemotherapy has not been conclusively shown to be of benefit for ependymomas, including grade 3 tumours⁹¹, it is still often tried in children with grade 3 tumours or tumours with negative prognostic molecular features.

Although myxopapillary ependymomas are WHO grade 2 instead of grade 1 in the new scheme, adult patients can simply be observed if the tumour has been completely excised^{46,92}. However, if the tumour cannot be completely excised, adjuvant radiation therapy is usually recommended^{46,93}. Treatment of paediatric myxopapillary ependymomas is more complicated, given the hesitancy to utilize craniospinal radiation in these patients, even in incompletely resected tumours that might eventually disseminate⁹⁴.

Embryonal tumours

Pathological features of embryonal tumours. The quintessential CNS embryonal tumour is medulloblastoma. Both the 2016 and 2021 WHO classifications of these tumours include four groups: WNT-activated; SHH-activated, *TP53*-wildtype; SHH-activated, *TP53*-mutant; and non-WNT, non-SHH (BOX 1). This classification reflects the fact that SHH-driven medulloblastomas are much more aggressive if they also have *TP53* mutations^{95,96}. Although WNT-activated medulloblastomas have excellent prognoses in children (less so in adults)^{97,98}, this seems to be the result of better response to adjuvant therapy. Without such therapy, WNT-activated medulloblastomas are highly aggressive,

Table 2 | Recommended treatment for ependymomas

Ependymal tumour	WHO grade	Molecular subtypes	Post-resection treatment recommendation	
			Gross total resection	Subtotal resection
Supratentorial ependymoma	2, 3	ZFTA fusion YAP1 fusion	Radiographic surveillance for grade 2; radiation therapy for grade 3	If subtotal resection or grade 3: age <1 year: adjuvant chemotherapy with cisplatin or carboplatin, cyclophosphamide and etoposide age 1–21 years: adjuvant radiation with chemotherapy age >21 years: adjuvant radiation or observation If disseminated disease: craniospinal radiation therapy
Posterior fossa ependymoma	2, 3	PFA	Adjuvant radiation therapy with chemotherapy	
		PFB	Observation	Adjuvant radiation therapy
Spinal ependymoma	2, 3	Spinal ependymoma Spinal ependymoma, MYCN-amplified	Observation	Focal radiation therapy
Myxopapillary ependymoma	2	NA	Observation	Focal radiation therapy
Subependymoma	1	NA	Observation	If subtotal resection and ongoing symptoms: observation or consideration for adjuvant radiation therapy

NA, not applicable^{127,128}.

and are thus still WHO grade 4. In a more recent study, non-WNT, non-SHH medulloblastomas were split into eight molecular subgroups on the basis of DNA methylation profiling, with subgroups 2 and 3 having the worst outcomes⁹⁹.

As for other CNS embryonal tumours, new entities in the 2021 scheme include ‘cribriform neuroepithelial tumour’, ‘CNS neuroblastoma, *FOXR2*-activated’, and ‘CNS tumour with *BCOR* internal tandem duplication’^{1,2}. The cribriform neuroepithelial tumour is in the 2021 edition as a provisional entity that is periventricular and has inactivation of *SMARCB1*. Thus, like atypical teratoid–rhabdoid tumours (ATRTs), cribriform neuroepithelial tumours show loss of nuclear INI1 by immunohistochemistry. Unlike ATRTs, they lack rhabdoid features and are arranged in strands and ribbons of cells, and therefore look more like choroid plexus carcinomas (CPCs). Further separating these tumours from ATRTs and CPCs is their relatively favourable response to therapy, with an average overall survival of 10 years¹⁰⁰. In contrast, the prognosis of *FOXR2*-activated CNS neuroblastomas is unknown because these tumours are so rare, although histologically they look highly malignant¹⁰¹ (Supplementary Fig. 6a,b). Previously, *FOXR2*-activated CNS neuroblastomas were often lumped into the non-specific ‘primitive neuroectodermal tumour (PNET)’ category, but so-called PNETs consistently segregate into other entities (most commonly glioblastoma) with advanced molecular testing¹⁰², and thus the PNET term is now obsolete in the 2021 WHO scheme. CNS tumour with *BCOR* internal tandem duplication is very difficult to diagnose solely on the basis of light microscopy findings, as it has perivascular pseudorosettes like an ependymoma (Supplementary Fig. 6c), as well as palisading necrosis like a glioblastoma. As suggested by the name, detection of a tandem duplication of the *BCOR* gene is required to make this diagnosis. Prognosis is variable — many patients experience a fast recurrence and short overall survival, whereas other patients do much better¹⁰³.

Clinical aspects of embryonal tumours. Clinical trials performed over the past decade have already resulted in separate therapeutic approaches for WNT-activated medulloblastomas versus other forms of medulloblastoma (Supplementary Table 3)¹⁰⁴. Additional studies are underway, including a single-arm study in children and young adults with WNT-activated medulloblastoma. In this study, participants are treated with lower doses of craniospinal radiotherapy (1,800 cGy) and less aggressive chemotherapy than the conventional approach, and patient outcomes will be compared with those in a historical control group^{96,105–107}. Whether the excellent prognosis seen in WNT-activated tumours will be maintained in studies of decreased intensity treatment is not yet known. The older terms ‘group 3’ and ‘group 4’, which were based on different mRNA expression patterns¹⁰⁸, are still often used in the context of non-WNT, non-SHH medulloblastomas; many clinical practitioners continue to identify group 3 *MYC*-amplified tumours as a distinct entity with a poor prognosis. There is general agreement that the classification scheme for medulloblastoma

remains a work in progress, as no uniform consensus about which subgroups of tumours should receive either more or less aggressive adjuvant therapy has been reached. Indeed, other than the WNT tumours in paediatric patients, patients with so-called average or good risk medulloblastoma still have a 15–25% chance of relapse, and the vast majority of patients who relapse ultimately succumb to their disease¹⁰⁹.

Standard therapy for medulloblastoma — other than the reduction in intensity for WNT-driven tumours in children and young adults, described above — remains relatively unchanged following publication of the 2021 WHO scheme⁵⁹. For children with average-risk disease, which continues to be defined as those without dissemination at the time of diagnosis who have undergone a complete or near complete resection (arbitrarily less than 1.5 cm² of residual disease) and do not show extensive histological evidence of anaplasia, treatment is begun with craniospinal radiation therapy (2,340 cGy) supplemented with boost radiotherapy to the tumour site¹⁰⁹. A variety of different chemotherapy regimens have been used both during and following radiation therapy^{96,109,110}. These chemotherapeutic approaches usually combine alkylators, such as cyclophosphamide and/or lomustine, with cisplatin and vincristine. For those patients with high-risk or poor-risk medulloblastoma, higher doses of craniospinal radiation therapy (3,600 cGy) are often coupled with more intensive chemotherapy, including high-dose chemotherapy supported with stem cell transplantation⁹⁶. Evidence from a randomized clinical trial in patients aged 3–21 years indicates that use of carboplatin during radiation therapy, primarily as a radiosensitizer, might provide a survival advantage in those with high-risk *MYC*-amplified tumours¹¹¹. Medulloblastoma arising in infants is difficult to treat, although SHH-driven infant tumours are often successfully managed with either aggressive chemotherapeutic regimens that use high-dose chemotherapy and stem cell transplantation, or systemic chemotherapy with concomitant intrathecal therapy^{111–113}. Even with similar treatment, survival is poor in infants with non-WNT, non-SHH infant medulloblastoma, as only 30–40% of patients are disease-free 3 years after diagnosis^{111,114–116}.

The other CNS embryonal tumours remain a highly problematic group of neoplasms therapeutically. CNS tumours with *BCOR* internal tandem duplications, as well as *FOXR2*-activated CNS neuroblastomas, are very difficult to manage. New molecular approaches have not yet been identified for either tumour. Separation of these tumours into molecular subsets is still critical for the development of better therapies, but the number of patients eligible for targeted treatments remains very small. Such clinical trials require not only national but international cooperation to be plausible^{34,117}.

Meningiomas

Pathological features of meningiomas. The genetic profiles of meningiomas have been the focus of a great deal of work between the 2016 and 2021 editions of the WHO classification. This work has resulted in the association of specific mutations with particular morphological subtypes or locations (for example, *BAP1* mutations in

rhabdoid meningiomas, *SMO* mutations in olfactory groove meningiomas, and *KLF4* and *TRAF7* mutations in secretory meningiomas). However, unlike the gliomas and embryonal tumours discussed above, none of the gene-specific molecular alterations drive prognosis, with two exceptions — *TERT* promoter mutations and homozygous *CDKN2A/B* deletion^{118,119}. When either of these lesions is encountered, the meningioma is grade 3, even if it does not quite meet the histopathological criteria for a grade 3 meningioma^{1,2}. Although most individual gene alterations do not seem to be prognostically important, evidence indicates that larger chromosomal alterations (such as chromosome 1p loss) and methylation subclusters might provide better outcome stratification when incorporated into multivariable algorithms alongside histopathology, *TERT* and *CDKN2A/B*^{118,120–124}. Another important change in the 2021 WHO scheme is that meningiomas that have rhabdoid or papillary patterns by histology are no longer automatically considered WHO grade 3 tumours, as long as other histopathological and/or molecular features of grade 3 tumours are absent¹.

Clinical aspects of meningiomas. From a clinical perspective, the 2021 WHO classification scheme does not substantially change how meningiomas are managed. Patients with complete resection of grade 1 and grade 2 meningiomas can be observed, although adjuvant radiation therapy is an option for patients with grade 2 meningiomas^{46,125}. Whether grade 1 and grade 2 meningiomas that are completely resected but have molecular alterations associated with worse prognosis (such as specific chromosomal copy number alterations) should be treated with adjuvant radiation therapy is an evolving area of research that requires more definitive studies^{120,121}. Patients with incompletely resected grade 2 meningioma, and all patients with grade 3 meningioma, require postoperative radiation therapy¹²⁵. This includes meningiomas that appear low-grade when assessed by light microscopy but have a *TERT* promoter mutation and/or *CDKN2A/B* deletion. However, as described above, emerging data suggest that more extensive revisions to the classification of meningiomas are likely to occur in the next edition of the WHO scheme, informed by ongoing clinical trials such as RTOG 0539 (REF.¹²⁶). Future revisions to meningioma classification should help determine which patients require more aggressive

postsurgical adjuvant therapy, as well as select patients for specific clinical trials.

Conclusions

The 2021 WHO classification of CNS tumours represents a major advance that has important implications for the treatment of patients with brain tumours. This new classification will improve the accuracy of diagnosis, provide better guidance on prognosis, allow selection of more appropriate therapies, and facilitate the enrolment of more homogeneous patient populations into clinical trials. However, it also introduces some challenges. For example, whether the biology and natural history of molecular glioblastoma are always and fully equivalent to histological glioblastoma, and whether both subtypes should be enrolled into glioblastoma clinical trials, remain unclear. Whether the treatment of grade 4 IDH-mutant astrocytoma should be similar to the treatment of IDH-wildtype glioblastoma is also not yet known. The introduction of many new types of brain tumour in the 2021 classification, especially in the paediatric population, enables selection of more specific therapies. However, this increase in biological specificity comes at the expense of clinical trial feasibility, as the size of eligible patient cohorts shrinks. Innovative strategies for the less common CNS tumour types and subtypes, such as the expansion of international cooperative groups, will be required to compensate. Furthermore, the new classification scheme puts increased pressure on pathology laboratories for faster, upfront advanced molecular testing. Even with faster turnaround, patients and clinicians will often need to wait longer than they previously did for a definitive diagnosis and treatment selection, including clinical trial enrolment. Third party payors might question the utility of this molecular testing, given the relatively few proven treatment options available for many CNS tumours. In such circumstances, it is important to emphasize that molecular testing also has great utility in confirming and refining diagnoses and prognoses, in addition to suggesting targeted treatments and fostering more therapeutic innovation. Despite these challenges, the 2021 WHO classification of CNS tumours will improve patient care, and will encourage the development of more effective therapies.

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A landmark study that demonstrated the ability of whole-genomic DNA methylation profiling to classify CNS tumours, including those that are difficult to classify by traditional microscopy and next-generation sequencing.

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